

Lab 4

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```
library(SDaA)
```

```
## Warning: package 'SDaA' was built under R version 4.2.3
```

```
data(coots)  
attach(coots)
```

4. One Stage Cluster Sampling Selecting a random sample of two clusters:

```
##clusterd<-1:100  
##sample(clusterd,2)  
## [1] 50 5  
## Commenting out input after initial knit to prevent new numbers ##being generated each time because I
```

square #5: 8x5=40 square #50: 10x5=50 N=100, n=2

Calculate the estimated mean, the standard error, and the estimated total:

Cluster #5:

$$M_i = 40$$

$$\bar{y}_i = 40$$

$$t_i = 40$$

Cluster #50:

$$M_i = 50$$

$$\bar{y}_i = 50$$

$$t_i = 50$$

$$\begin{aligned}\hat{\bar{y}}_r &= \Sigma M_i \bar{y}_i / \Sigma M_i \\ &= 40(40) + 50(50) / 90 \\ &= 45.556\end{aligned}$$

$$\bar{M} = 45$$

$$M_i^2 (\bar{y} - \hat{\bar{y}}_r)^2$$

$$\text{Cluster 5} = 40^2 (40 - 45.556)^2 = 51257$$

$$\text{Cluster 50} = 50^2 (50 - 45.556)^2 = 47089$$

$$\Sigma M_i^2 (\bar{y} - \hat{\bar{y}}_r)^2 = 98346$$

$$SE = \sqrt{(1 - (2/100)) (1 / ((2)(45)^2)) (98346 / 1)} \approx 4.878$$

$$N/n = 100/2 = 50$$

$$\hat{t}_{unb}$$

$$=50(90)=4500$$

Estimated Mean= 45.66

Standard Error= 4.878

Estimated Total= 4500

All equations used reflect the fact that the clusters are of uneven sizes, 5.16, 5.17, and 5.12 were used (in order)

Evidently I am supposed to have an $n=20$ not $n=2$ according to one of the problems below. So the above is for $n=2$, I will add the work for $n=20$ below.

```
clusterd<-1:10  
sample(clusterd,2)
```

```
## [1] 6 10
```

```
clusterd2<-11:20  
sample(clusterd2,2)
```

```
## [1] 15 20
```

```
clusterd3<-21:30  
sample(clusterd3,2)
```

```
## [1] 23 26
```

```
clusterd4<-31:40  
sample(clusterd4,2)
```

```
## [1] 39 33
```

```
clusterd5<-41:50  
sample(clusterd5,2)
```

```
## [1] 48 44
```

```
clusterd6<-51:60  
sample(clusterd6,2)
```

```
## [1] 54 52
```

```
clusterd7<-61:70  
sample(clusterd7,2)
```

```
## [1] 63 62
```

```
clusterd8<-71:80  
sample(clusterd8,2)
```

```
## [1] 77 80
```

```
clusterd9<-81:90  
sample(clusterd9,2)
```

```
## [1] 87 88
```

```
clusterd10<-91:100  
sample(clusterd10,2)
```

```
## [1] 97 93
```

N=100, n=20

Cluster1: 8,1
Cluster2: 16,18
Cluster3: 27,29
Cluster4: 39,35
cluster5: 43,48
Cluster6: 60,59
Cluster7: 66,62
Cluster8: 79,76
Cluster9: 87,86
Cluster10: 93,97

Area 8: 28
Area 1: 12
Area 16: 24
Area 18: 56
Area 27: 24
Area 29: 90
Area 39: 30
Area 35: 35
Area 43: 12
Area 48: 54
Area 60: 25
Area 59: 42
Area 66: 56
Area 62: 48
Area 79: 12
Area 76: 27
Area 87: 48
Area 86: 12
Area 93: 1
Area 97: 42

Calculate the estimated mean, the standard error, and the estimated total

$$\hat{y}_r = \Sigma M_i \bar{y}_i / \Sigma M_i$$

```
area.t<-c(28,12,24,56,24,90,30,35,12,54,25,42,56,48,12,27,48,12,1,42)
y.bar<-(sum(area.t))/20
y.bar
```

```
## [1] 33.9
```

```
M<-sum(area.t)
(sum(M*y.bar))/sum(M)
```

```
## [1] 33.9
```

$$\hat{y}_r = 33.9$$

$$\bar{M} = 33.9$$

$$M_i^2(\bar{y}_i - \hat{y}_r)^2$$

Cluster 1: $28+12=40/2=20$

Cluster 2: $24+56=80/2=40$

Cluster 3: $24+90=114/2=57$

Cluster 4: $30+35=65/2=32.5$

Cluster 5: $12+54=66/2=33$

Cluster 6: $25+42=67/2=33.5$

Cluster 7: $56+48=104/2=52$

Cluster 8: $12+27=39/2=19.5$

Cluster 9: $48+12=60/2=30$

Cluster 10: $1+42=43/2=21.5$

```
M.obs<-c(20,40,57,32.5,33,33.5,52,19.5,30,21.5)
M.bar<-(sum(M.obs))/10
M.bar
```

```
## [1] 33.9
```

```
M1<-33.9^2*((20-33.9)^2)
M2<-33.9^2*((40-33.9)^2)
M3<-33.9^2*((57-33.9)^2)
M4<-33.9^2*((32.5-33.9)^2)
M5<-33.9^2*((33-33.9)^2)
M6<-33.9^2*((33.5-33.9)^2)
M7<-33.9^2*((52-33.9)^2)
M8<-33.9^2*((19.5-33.9)^2)
M9<-33.9^2*((30-33.9)^2)
M10<-33.9^2*((21.5-33.9)^2)
M.need<-sum(M1+M2+M3+M4+M5+M6+M7+M8+M9+M10)
M.need
```

```
## [1] 1690373
```

```
sqrt((1-(20/100))*(1/((20)*(33.9^2)))*(1690373/(20-1)))
```

```
## [1] 1.759725
```

Standard Error=1.759725

$\hat{t}_{unb} = N/n = 100/20 = 5$

```
5*(sum(area.t))
```

```
## [1] 3390
```

$\hat{\bar{y}}_r = 33.9$

SE=1.759725

Total Estimate=3390

From lab 2 had the following information:

$\bar{y} = 34.1$

Total Area=2440

The estimates for the average value are fairly close to one another, however the original estimate for total area found in lab 2 was far from the true value of 3079. But, the estimate we have found using the clusters was much closer at 3390. While not exact, it is more accurate than the value found in lab 2.

5. Point Estimation

So do I need to find a confidence interval for this question or is it a question?

6. The point estimator for totals and proportions:

What other information would you need in order to estimate the total yield of the whole population of 150 farms?

We would want an measure of the Mean Square Within and Between to make sure the difference between the measurements obtained wasn't so great that we would be unable to make a determination from it. We would also want to know the ICC to give us an idea of the homogeneity within the clusters. This would also let us know if the sample size we have selected will yield a high enough precision to be useful, or whether we should consider a higher sample size or a different testing method altogether such as SRS. Additionally, is an acre a large enough size for each measurement to be able to accurately produce a usable estimate?

How would you obtain from equation 5.15 a point estimator for a population proportion when the variable of interest is categorical (yes/no) type?

For non categorical data (quantitative) we can use unbiased estimators to calculate a point estimator for the population proportion. However, when we are unable to use unbiased estimators, such as in cases like this question where the variable of interest is categorical we can use Ratio Estimation to estimate what proportion of responses were yes or no.

7. So am I supposed to find a confidence interval both here and for 5 or this supposed to be finding confidence intervals with two different methods or?

8. Example 5.6 from the book

```
M<-c(20,26,24,34,26,28,19,32,17,21,26,26)
ybar<-c(61.5,64.2,58.4,58,58,64.9,55.2,72.1,58.2,66.6,62.3,67.2)
```

Write the necessary R code to estimate the mean score in the population (equation 5.15)

$$\hat{\bar{y}}_r = \Sigma M_i \bar{y}_i / \Sigma M_i$$

```
sum(ybar*M)/sum(M)
```

```
## [1] 62.57191
```

```
#[1] 62.57191
```

Write the necessary R code to calculate the standard error (equation 5.17)

```
sqrt((1-(12/187))*(1/((12)*(24.92^2)))*(194827/(12-1)))
```

```
## [1] 1.491378
```

```
#[1] 1.491378
```

Code provides same answer as the book

9. Two Stage Clustering Sample

10.

11. The egg example

The example on pages 186-188 is that of the mean volume of eggs. The clusters are clutches of eggs and the individuals or secondary sampling units are the eggs. The variable of interest is volume. Assume you are given the following information:

of eggs in each clutch

Mean volume of the eggs in the sample for each clutch

Standard deviation in each sample

What R commands would allow you to do the calculations on page 188

```
Mi<-csize
y.bar<-tapply(coots$volume,coots$clutch,mean)
n.csize<-csize[seq(1,length(csize),2)]
t.est<-n.csize*y.bar
t.tot<-sum(t.est)
Mi.s<-sum(n.csize)
```

To find $\hat{y}_r$

```
y.est<-t.tot/Mi.s
y.est
```

```
## [1] 2.490579
```

$\hat{y}_r = 2.4906$

To find s_r^2 :

```
(1/(184-1))*sum((t.tot-(Mi*y.est))^2)
```

```
## [1] 38089480
```